

WELL COMPLETION REPORT (SYNTHETIC)

OIL-LKW-005 - Lakwa-005

NWIS prototype document - not an Oil India Limited operational record

API Number	OIL-ASM-000052
Field / District	Lakwa / Charaideo
Block	PEL-ASSAM-04
Basin	Assam-Arakan
Formation	Tipam (TIP)
Spud / Completion	2018-05-19 / 2018-11-06
Measured / TVD	2517 m / 2482 m
Trajectory	Vertical; inc 1.4 deg; az 316.3 deg
Formation Interval	1468-2264 m
Lithology	Sandstone-siltstone sequence
Pore Pressure Reference	2768 psi
Fracture Gradient	14.27 ppg
Mud System	KCl Polymer WBM
NWIS Well Risk	Moderate (50.3/100)

Casing Program

String	Set Depth (m)	Purpose
20 in conductor	80	Conductor
13 3/8 in casing	629	Surface
9 5/8 in casing	1459	Intermediate
7 in liner/casing	2214	Production

Significant Historical Events and Lessons

Date / Depth	Event	Severity	Evidence / mitigation / lesson
2018-07-10 896 m	Cementing Issue	Critical	Cement density/returns trend indicated placement uncertainty. Mitigation: Use staged cementing response when returns deteriorate. Lesson: Plan conservative displacement rate in weak formations.
2018-09-09 1748 m	Mud Loss	Critical	Active pit volume declined during stable pump rate. Mitigation: Stage circulation and monitor pit volume before resuming drilling. Lesson: Track flow balance and pit volume trend before losses become severe.
2018-06-16 542 m	Mud Loss	High	Intermittent circulation losses developed while drilling ahead. Mitigation: Optimize hydraulics and maintain conservative ROP through loss interval. Lesson: Track flow balance and pit volume trend before losses become severe.
2018-08-10 1332 m	Kick	High	Drilling break followed by positive flow check. Mitigation: Increase mud weight within fracture-margin limits and monitor gas trend. Lesson: Escalate connection-gas trends before entering correlated pressure zone.
2018-08-25 1538 m	Pressure Spike	High	Short-duration high-pressure excursion was recorded. Mitigation: Reduce pump rate, verify restriction and restore circulation gradually. Lesson: Use depth-matched offset hydraulic baseline for early warning.
2018-09-04 1670 m	Kick	High	Unexpected background gas rise preceded influx. Mitigation: Maintain trip margin and use conservative connection practices. Lesson: Escalate connection-gas trends before entering correlated pressure zone.
2018-09-09 1740 m	Kick	High	Flow-out exceeded flow-in and gas units increased. Mitigation: Re-establish overbalance and monitor pit/flow balance. Lesson: Treat pit gain plus gas increase as an early influx indicator.

Completion Summary

The synthetic well reached 2517 m MD in the Tipam interval. The NWIS knowledge record links 18 historical operational events to depth-matched telemetry. Risk classification is derived from formation, depth, trajectory and event history rather than formation alone.

This report is generated solely for SIH/NWIS prototyping and model development.